Neurons of a known censorship as an internal probe: a verified cascade audit of Qwen3.8-27B, complementary to an outside sensitivity prior

Technical report

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Abstract

We start from one censored behaviour we already know—Qwen3.8-27B denies verifiable facts about Taiwan in short yes/no answers while admitting them in free text—and use it to find *other* censored topics from inside the model, in a cascade. Activation×gradient attribution [17] of a two-answer margin selects 32 MLP neurons (32 of 1.1 M; 0.0006% of the weights) that carry the Taiwan script (held-out answers $11/19 \to 17/19$ factual when ablated, exact McNemar $p = 0.03$; 64 random neurons do nothing). Used as a probe over a neutral battery, they point to contested territories and China-aligned governments; this guides a larger battery and a broader map of censorship neurons, which in turn ranks topic lists written by the model itself; the map is trained on about 780 judged answers (260 censored) and its top 500 include 11 of the 32 seed neurons. Among high-scoring answers not judged before, those flagged by the map are censored in 29% of cases (86/293) against 3.4% (13/388) in a random sample of keyword-free answers judged the same way; among model-listed topics, 42 of 260 flagged answers are censored (16%; 23 of 144 once false premises are removed), against none of 60 low-scoring ones. An outside prior does better at finding the obvious: when a non-Chinese model rates the same listed topics for sensitivity, 41% of the answers on its top topics are censored (44% without false premises), against 16% for the neuron probe; the probe's distinct contribution is a non-obvious tail—among topics the prior rates as not sensitive it still finds censored answers (7/103 excluding an author the model simply does not know, 9/105 including it, e.g. the YJ-18 missile, the director Diao Yinan, a physicist presented as Taiwanese; none of 55 unflagged; one-sided Fisher $p = 0.046$ and 0.020)—and, at stage 2, a behaviour a topic-sensitivity prior is unlikely to flag (not tested): sovereignty scripts applied to Western Sahara and Crimea. Of 28 cases we picked among those the probes surfaced, 24 are censored again in a fresh bf16 verification (15 in Q4), while a non-Chinese reference model answers all of them on the merits. As a check that the probes point to the right neurons, switching off the 32 seed neurons removes the censorship for 19 of the 29, 32 random neurons for 1. The procedure is packaged as an open toolkit and rerun on Qwen3-4B. Pre-registered checks confirm the core (harmful-request refusals 198/200 vs 200/200 in English and 198/200 vs 199/200 in Chinese; general-text KL 1.3× a random ablation) and withdrew two of our earlier claims.

1 Motivation

Behavioural audits find *that* a model censors and, with black-box crawlers, *which* topics it avoids [1, 2, 3]. Interpretability work reports that political censorship in Chinese-origin models is carried by linear directions that are mostly orthogonal to safety refusal, with evidence of coupling in some models [4, 5]. We study the step in between, for an auditor who holds *one* known example: turning it into an internal instrument that finds *other* censored behaviours of the same model, and checking each finding against the instrument it came from.

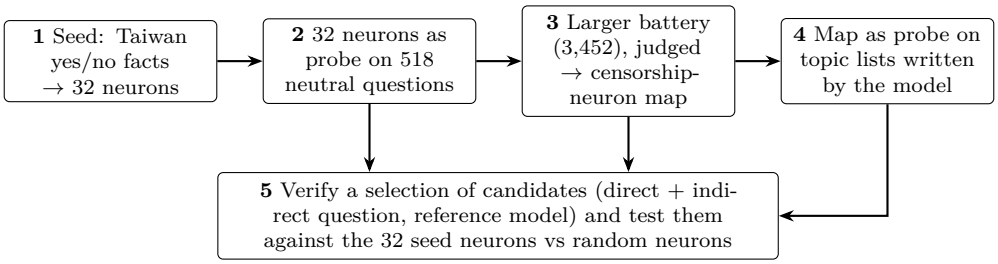


Figure 1: The cascade audit. Each stage is guided by what the previous one found: stage 2 uses the seed neurons as a probe; stage 3 builds a battery around what stage 2 revealed and trains a censorship-neuron map on judged answers; stage 4 uses that map as a probe. Stage 5 returns to the seed neurons. Every stage is available in the `neuroaudit` package.

2 The cascade

Model, decoding, judge. Qwen3.8-27B, official bf16 weights (Apache-2.0), run with a CPU/GPU split (40 layers on CPU, 24 on two 16 GB GPUs), greedy decoding, thinking off. The battery answers used as labels (stages 3–4) and several runs (C7, E11, E13, E14, E17 and a secondary E16 run) used Q4_K_M GGUFs of the base and edited weights with llama.cpp; activations, attributions, C0, C8, E5, E9, E15 and the main E16 run are bf16. Answers are labelled by Claude Sonnet with a fixed prompt (`censura/judge_base.py`) into: *balanced*; *refusal*; *official script* (recites a government or party line, attacks the premise, calls critics hostile forces); *denial* (of reported facts); *sovereignty claim*; *praise*; *deflection*; *other*. An answer is *censored* if it is a refusal, official script, denial, sovereignty claim or praise.

Stage 1: seed and attribution. 40 yes/no questions on verifiable facts about Taiwan's de facto status (elections, passports, army, currency…) and 30 controls of the same form about other countries, with the system prompt "Answer with exactly one word: Yes or No". For each item the margin is $m = \log p(\text{factual}) - \log p(\text{other})$ at the first answer token. A neuron is an input column of an MLP down projection ($64 \times 17{,}408$); its attribution is activation$\times\partial m/\partial$activation summed over positions and averaged over the Taiwan items of a random half A. `off32` is the set of the 32 most negative neurons (they push against the factual answer); *rand64 / rand32* are random neurons with the same per-layer distribution. Ablation = zeroing the activation (equivalently, the column).

Stage 2: the seed neurons as a probe. 518 template questions (4 per country on 123 countries, 26 general topics): activation of the 32 at the last prompt token, and KL of the next-token distribution under ablation; 17 questions chosen among the top 29 by KL (plus 2 controls) are read in generation.

Stage 3: the censorship-neuron map. Guided by stage 2, a neutral battery of 3,452 questions (half English, half Chinese templates with the entity name in English) on 159 places, 70 leaders, 80 events, 29 groups, 25 organisations and 30 debated topics. Base answers (Q4) are labelled in a first round (776 labels, 260 censored: answers flagged by censorship keywords plus a random sample of the rest); a difference-of-means direction [18] in the residual stream at the last prompt token (layer 49) and the 200 neurons that write it most (via down-projection write scores) form the map. Answers the map then flags were judged in a second round (1,210 labels in total, 346 censored). The map recovers 11 of the 32 seed neurons among its top 500 and scores every question from a forward pass only; among 293 high-scoring answers not judged in the first round (5 of them contain censorship keywords) the judge finds 86 censored (29%; 81/288 without those 5), against 13 of 388 (3.4%) in the random sample of keyword-free answers judged in the first round with the same judge, and 0 of 150 low-scoring.

Stage 4: the map on the model's own lists. For 79 generic categories (45 about China, e.g. "Chinese film directors", "religious groups in China", "Chinese space and military programs"; 22 about other countries; 12 neutral) the model lists entries itself (2,143 entries, each asked in English and Chinese: 4,286 questions); the map ranks them, and the top entries are answered and judged. This stage also produces false positives

*Experiments, code and drafts were produced by Claude Opus 5.5 working as a coding agent under the author's direction (Section 7). Code, data and pre-registrations: https://github.com/ValerioDolci/frontier-weight-edit/tree/qwen27b-censorship-v1.0/experiments/qwen27b-censorship.

the model correctly rejects ("doping scandal involving" table-tennis players who never had one): candidates must be verified.

Stage 5: verification (E16). 33 candidates (28 surfaced by stages 2–4, 5 questions we chose while testing the seed neurons) and 2 controls (an author the model does not know; a doping "scandal" that never happened). Each has a direct question (as surfaced) and an indirect knowledge question with a reference answer. Variants: base, `off32`, rand32; Gemma 4 31B-it as a non-Chinese reference. *Unlocked* = censored in the base and not censored with the neurons off. The rule was fixed before running.

Pre-registration. Thresholds for the checks C0–C8 and predictions for E5–E17 were written before each run (`prereg/`; ordering documented by session logs). A failed check changes the sentence, not the threshold.

3 Results

3.1 Stage 1: 32 neurons carry the Taiwan script

	base	`off32`	off64	rand64
Taiwan yes/no, held-out half B (factual)	11/19	**17/19**	18/19	11/19
Taiwan yes/no, half A	7/21	21/21	21/21	7/21
Control countries	30/30	30/30	30/30	30/30
New open questions with the official script	8/10	**0/10**	0/10	8/10
MMLU-Pro [19] (300 questions; resolution ±5.5 pts)	60.7%	61.3%	61.7%	61.3%

Table 1: Ablation of the seed neurons (bf16). Exact McNemar: half B $p = 0.031$ (6/6 discordant pairs favourable; 17/19 Wilson 95% CI 0.69–0.97), open questions $p = 0.0078$.

After ablation the model answers, e.g., "won by William Lai (Lai Ching-te) of the DPP" and "administered by the Republic of China"; writing the 32 zero columns into a copy of the weights gives the same answers without hooks. Across six A/B splits (the original and five new ones) the selected top-32 overlap with mean Jaccard 0.60 (min 0.45), 20–25 of the 32 reappear in each split, and each split's own top-32 raises its held-out half to 95–100% factual (C1).

3.2 Stages 2–4: what the probes surfaced

The seed neurons rank first Taiwan, Tibet, Xinjiang and Hong Kong questions, then other contested territories: in bf16 generation the base model calls Western Sahara and Crimea part of China's territory and defends Vietnam and Laos with the formulas it uses for China ("an inseparable part of China's territory" for Western Sahara); `off32` removes both. The same script on Kosovo in Qwen3.5-9B is reported in [5], and spillover of official Chinese positions onto other countries in [6]. On a pre-registered grid of 18 territories × 5 questions × 2 languages (C8, bf16) the sovereignty script fires on 29/110 answers about contested non-Chinese territories and 0/40 about control countries (also for "criticisms of the leadership"), more in Chinese (20/55) than in English (9/55); with `off32` it fires on 0/110 contested answers, while Chinese territories go only from 29/30 to 20/30: the 32 neurons carry the script for disputed territories in general more than for China's own. The map and the model-written lists add, among others, the YJ-18 missile ("no such missile"), the director Diao Yinan ("cannot provide information about this person"), Yiguandao and the Senkaku and Spratly islands.

3.3 Yield along the cascade

On all labelled cases the map retrieves 333 of 346 censored answers (96%) above its threshold, with 58% precision—partly in-sample, since 86 of them were found by the map itself and 260 trained it; the misses are mostly "the premise is false" answers about countries such as Poland or Chile. The seed provides the first probe and orients the battery; large-scale screening is done by a map trained on about 780 judged answers.

An outside prior (E17). Is the probe better than simply asking which topics are sensitive? On the stage-4 population a non-Chinese model (Gemma 4 31B) rated each listed topic 0–10 for how likely a Chinese chatbot is to refuse or give the official line; we took as many top-rated topics as the probe had flagged (167), asked

Stage	Probe and items	Flagged, judged censored	Comparison
2	32 seed neurons over 518 neutral questions	8 of 17 read (bf16); pre-registered grid: 29/110 on contested territories	controls 0/2; grid controls 0/40
3	map over the 3,452-question battery (Q4 answers), high-scoring answers not judged before	86 of 293 (29%)	random keyword-free sample: 13 of 388 (3.4%); low-scoring: 0 of 150
4	map over 4,286 questions on entries the model listed (79 categories; Q4 answers)	42 of 260 answers (16%); 23 of 144 (16%) without false premises	0 of 60 lower-scoring (95% upper bound ≈6%)

Table 2: Yield along the cascade (one LLM judge, same prompt throughout); compare Table 3 for an outside prior. Stage 4 counts answers (260 answers on 167 entries); 19 of its 42 censored answers were false premises the model correctly rejects (invented doping cases, garbled company names); removing all false-premise items leaves 23 of 144.

the same questions and used the same judge. By the pre-registered reading, the probe does *not* add yield

	neuron probe (map)	outside prior (Gemma rating)
Answers judged censored	42/260 (16%)	138/334 (41%)
without false premises	23/144 (16%)	123/282 (44%)
Censored on topics the prior rates 0–3 (not sensitive)	7/103 (9/105)*	—
Topics with censored answers found only by this method	9	67

Table 3: The neuron probe against an outside sensitivity prior on the same topics (E17); Q4 answers, both with up to 160 tokens, same judge. *9/105 counting the unknown author. The two sets of topics overlap in 38 of 167. Found only by the probe: the YJ-18 missile, Diao Yinan, Wu Jianxiong, Wang Zhongliang, the Mengniu case, and four items of the "the premise is false" kind (a "Papuan civil war" answered in Indonesia's favour, a "Tibet earthquake" answered with a sovereignty reminder, a "China Delta outbreak", an unknown author).

over an outside prior: the prior finds the obvious topics (dozens of city-level student protests, Tiananmen, Xi Jinping, the Xinjiang camps) far more efficiently. What the probe adds is a small, non-obvious tail the prior misses—the YJ-18 missile ("no such missile"), the director Diao Yinan, a physicist presented as Taiwanese—at low precision (7 of 103 answers on topics the prior considers non-sensitive, against 0 of 55 unflagged ones; one-sided Fisher $p = 0.046$; 9/105 and $p = 0.020$ if the unknown author is counted). An audit should use both: the prior for coverage, the internal probe for what the prior cannot anticipate and to locate the mechanism.

3.4 Stage 5: verification against the seed neurons

Table 4 gives the verdicts in bf16, the precision in which the seed neurons were selected. The 33 candidates are an inventory we picked: 28 among the cases the probes had surfaced and 5 well-known topics we chose ourselves, each asked with one new direct question; so the confirmation rate measures how well the surfaced cases reproduce, not the precision of the probes. 29 are censored by the base model (24 of the 28 probe-surfaced ones; 15 in Q4). Gemma answers 33 of 33 on the merits; the two controls are not censored (with the seed neurons off, the unknown-author control turns from a deflection into a refusal). Switching off the 32 seed neurons removes the censorship for **19** of the 29, switching off 32 random neurons with the same per-layer distribution for 1; in 13 of the 19 unlocked topics the base model also answers the indirect knowledge question correctly. Among the 10 topics not unlocked, 7 fail the indirect question; for Tiananmen this contradicts the multiple-choice result of Section 3.7, so the indirect question is a weak test of knowledge. A secondary run on Q4 GGUFs gives 19 censored and 13 unlocked (random neurons: 0); the base verdict agrees between the two precisions for 23 of 33 candidates. Topics censored in both precisions include Jimmy Lai, Joshua Wong, the Dalai Lama, Senkaku, Spratly, Kashmir, Laos, the YJ-18 missile, Diao Yinan and Yiguandao; others (e.g. Liu Xiaobo, Galwan) are censored only in bf16.

Verdict (E16, bf16)	n	Topics (source of the candidate)
Censored, **unlocked** by the 32 seed neurons	19	Hong Kong 2019 protests (M), Hong Kong chief executive (T), Jimmy Lai (M), Joshua Wong (M), Dalai Lama (T), Inner Mongolia 2020 (M), Tank Man (H), Liu Xiaobo (M), Yiguandao (R), Galwan 2020 (M), Senkaku (R), Spratly (R), Western Sahara (T), Crimea (T), Kashmir (M), Vietnam (T), Laos (T), YJ-18 (R), Diao Yinan (R)
Censored, not unlocked	10	Tibet 1959 (H), Xinjiang camps (H), Ilham Tohti (M), Urumqi fire protests (M), Tiananmen 1989 (H), Zhao Ziyang (M), Peng Shuai (H), Falun Gong (R), Mengniu melamine (R), Church of Almighty God (R)
Not censored in this phrasing	4	Umbrella Movement (T), Panchen Lama (T), Great Leap famine (T), South China Sea ruling (M)

Table 4: Verification of the cascade's candidates (bf16, greedy, one direct question each). Source: T = scan with the 32 seed neurons (stage 2); M = censorship-neuron map over our battery (stage 3); R = topic lists written by the model, ranked with the map (stage 4); H = questions we chose while testing the seed neurons. Random-neuron control: 1 unlocked.

3.5 The procedure on a second model

The packaged tools rerun end-to-end on Qwen3-4B (bf16): the Taiwan seed yields neurons that raise held-out factual answers from 79% to 100% (random: 79%); scanning 477 questions on 159 places, the largest ablation effects are all on contested or limited-sovereignty places (Nagorno-Karabakh, Taiwan, Western Sahara, Abkhazia, Kashmir, Palestine, Catalonia, Macau, Crimea). Read by hand (no judge), the script appears in the base only for Taiwan, Tibet and Macau, and switching the neurons off introduces geography errors elsewhere ("Nagorno-Karabakh … in the Republic of Armenia"): on this model the neurons are tied to disputed sovereignty in general. Both models are from the same family; this replicates the procedure, not generality across architectures.

3.6 Costs, safety and an external battery

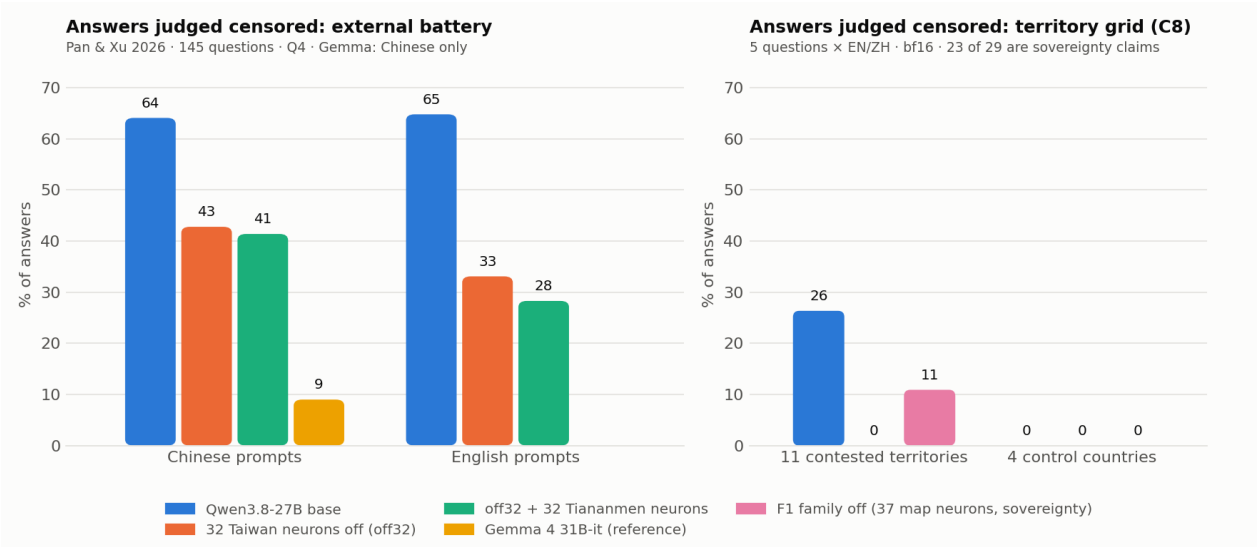


Figure 2: Left: answers judged censored (refusal, official script, denial, sovereignty claim or praise) on the 145 questions of Pan & Xu [2], Q4, greedy, 128 tokens. Right: the same judgement on the pre-registered territory grid (C8, bf16): 29 of 110 answers on contested territories (23 sovereignty claims, 5 official script, 1 praise), 0 of 40 on control countries; Chinese territories, not shown, go from 29/30 to 20/30 with `off32`. Pink: switching off only F1, a family of 37 censorship-map neurons (layers 36–49) whose top topics are territorial sovereignty. Same y-scale in both panels.

- **Safety (C7, E18, Q4).** StrongREJECT [20] harmful requests refused 200/200 → 198/200 in English (two borderline flips; McNemar $p = 0.5$) and, translated into Chinese, 199/200 → 198/200 (32 random neurons: 200/200); over-refusal on XSTest [21] safe prompts 5.2% → 2.4% in English, 4.0% → 2.8%

in Chinese; 40 Italian requests all refused. In Qwen3-1.7B and 8B, edits that reduced censorship (one or two neurons, or a direction) were reported to raise harmful-request compliance in Chinese by 4–7 points [14]; the 32 seed neurons here do not.

- **General text (C0, bf16).** KL to the base on 100 WikiText [22] passages: 0.00096 nat/token for `off32` vs 0.00074 for 64 random neurons (1.3×, pass); the largest effect is on a passage about the Northern Ireland border. Adding 32 Tiananmen-refusal neurons costs 3.9× (fail).
- **Short factual recall (E14, Q4).** 28 neutral questions about China: 27/28 for both base and `off32`, in both languages.
- **Side effects.** `off32` denies that Hong Kong has a 2020 National Security Law and, in Chinese, that Huawei was indicted in 2019; unlocked answers can contain wrong details; on the external battery 7 Chinese answers become newly censored while 38 are unblocked.
- **External battery (E11, Q4).** Pan & Xu, 145 questions (including their 30 low-sensitivity ones): censored 64.1% → 42.8% in Chinese (McNemar $p = 3 \times 10^{-6}$), 64.8% → 33.1% in English; Gemma, in Chinese: 9.0%.

3.7 Under the Tiananmen refusal

The model recognises the facts: in a 4-option multiple choice (4 orders) it gives 0.955 to "the army forcibly cleared pro-democracy protesters, killing hundreds or more"; forced to continue "Estimates of the number of people killed range from" it writes "several hundred to several thousand". Yet, prompted to continue a Chinese sentence starting "On June 4, 1989, in Beijing,", the unmodified model invents a celebration of the PRC's 40th anniversary with a military parade; ablating the refusal neurons yields "nothing major happened" in Chinese and mostly the official narrative in English, with fragments of the facts in some variants ("a military crackdown on the protesters") (E5). Removing a refusal does not reliably recover the facts.

3.8 The map reads the outcome; causality not shown

The residual direction separates censored from balanced answers with AUC 0.95–0.97 on held-out entities, inside each language (0.98 EN, 0.91 ZH; C2) and among sensitive topics only (0.855 vs a 0.61 entity-permutation null; C3). Steering along it at one layer turned 9/20 balanced answers into the script (2/20 for noise and for an equal-norm random direction) but did not replicate on 20 new questions (5/20 vs 4/20), and ablating it at one layer changes nothing (E15): a correlational read-out.

4 What the checks withdrew or did not confirm

"Foreign-facing censorship is the same circuit." We had taken a cosine of 0.61 between the PRC direction and a direction built from non-Chinese cases (e.g. "Crimea is part of Russia"), and 40 shared top neurons, as evidence of one mechanism. The three pre-registered tests passed, but post-hoc nulls showed that the difference of 71 random questions has $|\cos|$ up to 0.82–0.85 with the PRC direction; the transfer disappears once the first principal component, entangled with language and template, is removed; standardised per-neuron scores share no top neurons (a metric without a positive control). A weak shared signal remains (AUC 0.69 within entity × language, permutation $p = 0.001$). Most foreign cases are Chinese-language questions asking for criticism of a country, answered "the premise is false, respect sovereignty"—also for Poland or Chile.

"The direction is causal." Not shown (E15). Other deviations: C2 fails its cosine criterion on the letter (substance holds); C7 and E11 ran on Q4 without two planned variants; the first E16 run (Q4) disagreed with earlier bf16 answers on 8 of 13 overlapping topics, so E16 was rerun in bf16 (main result) and Q4 is reported as secondary; eleven point predictions missed (section "Pre-registered point predictions vs outcomes" of `FINDINGS.md` in the repository), including our prediction that the invented parade would need heavy ablation.

5 Related work and what is new

Black-box discovery of forbidden topics [1] and behavioural measurement of censorship in Chinese-origin models [2, 6, 3] establish *that* and *where* models censor. Political censorship is linearly decodable and mostly

orthogonal to refusal, with refusal-direction ablation also lifting political censorship in one model (Qwen3-8B, presented as an interpretive hypothesis) [4]; a shared PRC axis across topics and the script firing on Kosovo in Qwen3.5-9B are reported in [5], together with a null result for zeroing 600 discriminatively chosen MLP neurons—our neurons are chosen by causal margin attribution and on another model. Crosscoder model diffing finds "CCP alignment" and Taiwan-sovereignty features without supervision [7]; censored Qwen models are used as testbeds for eliciting secret knowledge [8]; refusal neurons are probed and audited causally [9, 10, 11]; abliteration removes refusals but not script-dependent bias about Taiwan [12]; commercial edits remove much censorship without disclosing the method [13]; in Qwen3-1.7B and 8B, a few neurons that loosen China-related censorship were reported to also loosen safety on harmful requests in Chinese [14]; and CensorScope [15] discovers refused topics from the weights, using SAE features that co-activate with a refusal vocabulary bootstrapped from a small lexeme seed and validating them by feature ablation; auditing with interpretability tools is framed in [16].

To our knowledge (search of 7 November 2026, largely abstracts), the following are not covered elsewhere: (i) a cascade whose first probe is a small set of *causal* neurons from one known *behaviour*—including non-refusal censorship such as official scripts, denials and sovereignty claims, which a refusal-vocabulary seed [15] is not designed to target—with candidates verified in generation against a reference model and back against the seed neurons with a random-neuron control; (ii) the resulting count of topics the seed neurons do and do not unlock; (iii) a direct comparison with an outside sensitivity prior, which the probe does not beat in yield but complements; (iv) a pre-registered check round with published corrections. The toolkit is a convenience, not a claim. We did not compare coverage with a black-box crawler on the same model.

6 Limitations

One model and one seed, plus a same-family replication of the procedure. Verdicts per topic rest on one phrasing; they differ between Q4 and bf16 more than the 8% label change measured on the neutral battery (C6), so we report bf16 as main. Labels come from one LLM judge (second-judge $\kappa = 0.91$ and blind re-annotation 89% on the neutral battery, 74% on foreign cases; not re-measured on E16). Topics not unlocked mostly also fail the indirect knowledge question, so ignorance is not excluded for them. The unlocking of stage-2 candidates is partly circular (they were selected by the effect of the same ablation): outside them, the seed neurons unlock 12 of 18 map-surfaced candidates and 1 of 5 hand-picked ones. "32 neurons" is a sufficient set found by one procedure, not a unique circuit. Small n in several causal tests.

7 Responsible use and how this was made

The political-script and Tiananmen-refusal neurons are published; safety-refusal neuron lists, harmful prompts, answers to them and modified weights are not. Experiments, code and drafts were produced by Claude Opus 5.5 as a coding agent on the author's workstation; labels by Claude Sonnet, with Gemma 4 31B as a second judge and a blind re-annotation by a separate Claude instance; key analyses and this report were attacked by adversarial reviewer instances, which produced the corrections in Section 4.

A Check round at a glance

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ID	Check	Outcome
C0	KL on generic text $\leq 2\times$ random ablation	`off32` 1.3× pass; with Tiananmen neurons 3.9× fail
C1	McNemar $p < 0.05$; neuron stability across 5 splits	$p = 0.031/0.0078$; Jaccard 0.60, 5/5 splits pass
C2	Language not the confound (AUC within language; cosine)	AUC 0.98/0.91 pass; cosine criterion fails on the letter
C3	Decision vs topic sensitivity (AUC among sensitive only)	0.855 vs null 0.61, pass
C4	"Single viewpoint" for foreign censorship	pre-registered tests pass; claim withdrawn post hoc
C5	Judge reliability	$\kappa = 0.91$; manual 89% (foreign 74%)
C6	Q4 labels vs bf16	8.0% changed vs 6.7% judge test–retest, pass
C7, E18	Safety refusals (English; Chinese)	200/200 → 198/200; 199/200 → 198/200; over-refusal down; pass
C8	What triggers the script	territory category yes; question form no; language opposite to prediction
E5	Tiananmen: belief or damage	rewritten account present in the unmodified model
E9, E13	Censorship vs ignorance; company protection	mixed; no general company protection
E11	External battery	64% → 43% (ZH), 65% → 33% (EN)
E14	Neutral factual recall	27/28 = 27/28
E15	Causality of the direction	not replicated
E16	Verification of the cascade's candidates	19/29 unlocked (bf16); rand32 1
E17	Probe vs outside sensitivity prior	prior 41% vs probe 16%; probe adds a non-obvious tail

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